

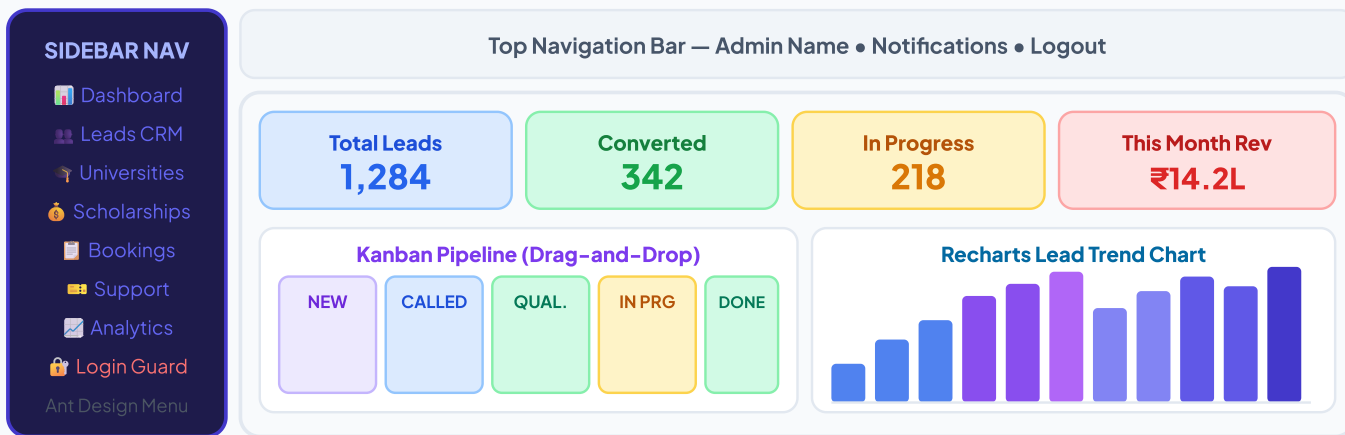
Admin CRM Operations Portal

Ant Design v5 data tables, drag-and-drop Kanban pipeline, Recharts analytics dashboard, Axios interceptors with auth loop guards, and crawler immunity.

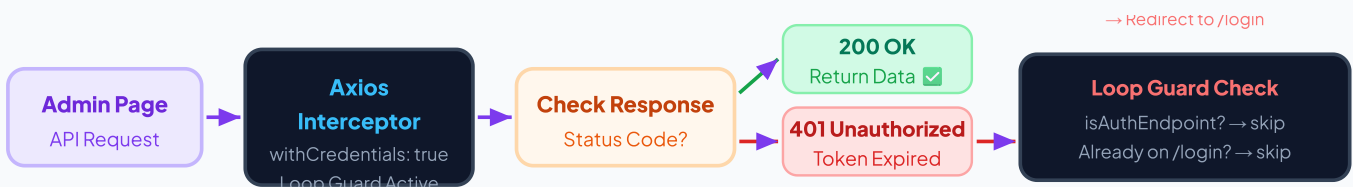
🎨 UI: Ant Design v5 📊 Charts: Recharts SVG 🛡️ Crawler-Immune: noindex headers

1. Admin Portal Architecture

ADMIN CRM PORTAL — PAGE LAYOUT & COMPONENT ARCHITECTURE



AXIOS AUTH INTERCEPTOR — JWT LOOP GUARD & AUTO-LOGOUT FLOW



2. Technology Decision Framework

1. Ant Design v5 vs Material UI vs Custom CSS

UI LIBRARY

WHAT IS IT?

A React enterprise UI component library providing pre-built data tables, forms, modals, date pickers, and layout components optimized for admin dashboards.

HOW DOES IT WORK & PROBLEM SOLVED?

Building a sortable, filterable data table from scratch for 1,000+ lead records would take 2–3 weeks. Ant Design's `<Table>` component provides server-side pagination, column sorting, row selection, and inline editing out-of-the-box, fully accessible and keyboard-navigable.

WHAT ALTERNATIVES EXISTED?

Material UI: Similar scale, but Google-branded aesthetic that conflicts with UniCoach's brand language. **Custom CSS:** Maximum flexibility but 10x development time.

WHY DID WE CHOOSE THIS APPROACH?

Ant Design's data-dense table components are purpose-built for CRM operations. Token customization via ConfigProvider lets us override the default blue palette to match UniCoach brand colors.

2. Crawler Immunity via Vercel Security Headers

SECURITY

WHAT IS IT?

A `vercel.json` configuration that injects `X-Robots-Tag: noindex, nofollow` and `X-Frame-Options: DENY` HTTP headers on all admin portal responses.

HOW DOES IT WORK & PROBLEM SOLVED?

Without these headers, Google could index the admin login page, student data dashboards, or internal lead records. Search engine crawlers would discover and cache sensitive operational data. These headers instruct all crawlers to exclude the admin domain from their indexes entirely.

WHAT ALTERNATIVES EXISTED?

robots.txt: Voluntarily respected but not enforced. Malicious crawlers ignore it. HTTP headers are enforced at the transport layer.

WHY DID WE CHOOSE THIS APPROACH?

Prevents sensitive operational data from appearing in Google search results and denies iframe embedding attacks (clickjacking).

💡 Senior Interview Defense: "How did you architect the admin CRM?"

"The UniCoach admin portal is a purpose-built CRM SPA using Ant Design v5 for its enterprise-grade data table components. The Kanban drag-and-drop pipeline uses `dnd-kit` under the hood, triggering `PATCH /api/leads/:id/status` API calls on drop. Recharts renders SVG lead trend charts. All API calls are routed through a centralized Axios interceptor with an infinite loop guard — if a 401 is received outside an auth endpoint or login page, the session is immediately cleared and admin is redirected to login. Finally, we deploy with `noindex/nofollow` headers to prevent search engines from indexing any admin UI or lead data."